
Bilinear C_4 -Covariant Spectral Neural Operator (BiC4-NO) for Scientific Field Surrogates and Gravitational Lensing

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Abstract

1 Neural operators provide efficient surrogates for scientific field mappings, but
2 standard spectral operators do not enforce how scalar and vector fields transform
3 under spatial rotations. This matters in gravitational lensing, where rotating a scalar
4 convergence field should correspondingly rotate the vector deflection field and the
5 leading nonlocal response is analytically known. We introduce BiC4-S3NO, a
6 physics-guided C_4 -covariant neural operator with radially shared spin-spectral fil-
7 ters, symmetry-compatible bilinear interactions, and task-specific physical adapters.
8 On strong-lensing simulations, BiC4-S3NO achieves the lowest relative L^2 er-
9 ror across held-out warm-dark-matter and zero-shot cold-dark-matter and axion
10 domains at two training-set sizes, while preserving quarter-turn covariance to
11 numerical precision. It also obtains the lowest development-validation error on
12 Navier–Stokes; on Darcy flow it ranks second at both native and zero-shot reso-
13 lutions. The results show that structured symmetry and analytic physical seeds
14 can improve scientific-field surrogates when they match the discretized problem,
15 while Darcy exposes limitations when grid and boundary geometry reduce the valid
16 symmetry.

17 1 Introduction

18 Neural operators learn maps between functions and can amortize the cost of repeatedly solving related
19 physical systems [1]. Fourier Neural Operators (FNOs) achieve global information exchange through
20 learned multipliers on a truncated Fourier grid [2]. Their unconstrained mode-wise weights, however,
21 do not encode relationships between frequencies connected by rotation. A scalar-to-vector map must
22 therefore learn both the physical relation and the transformation law from data.

23 This issue is central to gravitational lensing and the broader search for the nature of dark matter [3].
24 Dark-matter substructure perturbs lensed images and offers a probe of subgalactic structure [4–6];
25 modern lens-modelling software supports the associated simulation and inference workflows [7]. In
26 the task studied here, a scalar convergence field is mapped to a vector deflection field. The mapping
27 is nonlocal, its leading inverse-gradient response is known, and a quarter-turn of the input should
28 produce the same quarter-turn of the output vector. Existing operator variants address complementary
29 representation questions: PDNO learns a frequency-conditioned symbol [8], while LNO and ALNO
30 use Laplace-domain components intended primarily for transient responses [9, 10]. Group-equivariant
31 networks and FNOs constrain spatial symmetries by construction [11, 12], but do not incorporate the
32 spin-typed, physics-seeded residual construction used here.

33 We introduce the Bilinear C_4 -Equivariant Spin-Spectral Neural Operator (BiC4-S3NO). Its channels
34 carry prescribed rotational types; its Fourier multipliers share radial parameters while retaining the

angular harmonics allowed by the four rotations of a square grid; and sparse bilinear routes represent nonlinear interactions without violating those types. Fixed task adapters supply known physical responses and boundary treatments, leaving the neural operator to learn the discrepancy. We evaluate this design on strong-lensing simulations derived from the DeepLense benchmark and software lineage [13, 14], Darcy flow, and forced Navier–Stokes.

2 Method

2.1 Spin-spectral operator

A complex spin- s field transforms under a rotation R_β as

$$[\rho_s(R_\beta)z](\mathbf{x}) = e^{is\beta} z(R_\beta^{-1}\mathbf{x}), \quad \beta \in \{0, \pi/2, \pi, 3\pi/2\}. \quad (1)$$

Scalars have spin zero; a planar vector is packed as $u_x + iu_y$ and has spin one. For input spin s_i and output spin s_o , the learned Fourier multiplier is

$$W_{oi}(\ell) = \mathbf{1}_{\{|\ell| \leq \Lambda\}} \sum_{m=-M}^M g_{oim}(|\ell|) e^{i(s_o - s_i + 4m)\phi_\ell}, \quad \hat{y}_o = \sum_i W_{oi} \hat{x}_i. \quad (2)$$

Each complex radial profile g_{oim} is linearly interpolated from learned physical-frequency knots. The $m = 0$ component is continuously rotation-covariant, while the $4m$ sidebands add angular dependence without breaking C_4 covariance. Non-scalar DC paths and unpaired Nyquist modes are removed so every retained mode has the partners required by a lattice quarter-turn.

The core contains a spectral lift, four gated residual blocks, and a spectral projection. A block uses $u^\ell = S_\ell h^\ell$ followed by

$$h^{\ell+1} = h^\ell + \mathcal{P}[u^\ell \sigma(a^\ell |u^\ell| + b^\ell)], \quad (3)$$

where $\mathcal{P}$ restores rotation-paired support. Since $|u^\ell|$ is invariant, the gate preserves spin phase. Nonlinearity beyond channel-wise gating is supplied by sparse products $h_i h_j$ and $h_i \bar{h}_j$, instantiated only when their summed or differenced spins equal the output spin modulo four. Their coefficients are zero initialized. Navier–Stokes uses an independent bilinear module after every block and filters both factors and products to paired two-thirds spectral support.

2.2 Physics-aware adapters

For lensing, convergence κ , potential ψ , and deflection α satisfy [15]

$$\nabla^2 \psi = 2\kappa, \quad \alpha = \nabla \psi. \quad (4)$$

We compute periodic and symmetrically zero-padded discrete inverse-gradient predictions. The normalized output combines trainable multiples of these bases with a low-frequency spectral correction and the BiC4 residual,

$$\tilde{\alpha} = \gamma_{\text{per}} \tilde{\alpha}^{\text{per}} + \gamma_{\text{pad}} \tilde{\alpha}^{\text{pad}} + \mathcal{L}_{\text{low}}(\tilde{\kappa}) + \mathcal{N}_{\text{BiC4}}(\tilde{\kappa}). \quad (5)$$

The gains are initialized by training-only ridge regression, and learned corrections are initially zero. A full-band path covers all rotation-paired frequencies on the finite grid.

Darcy maps permeability and fixed coordinates to pressure with zero-Dirichlet conditions. Its half-open coordinate grid is not closed under continuum quarter-turns, so the end-to-end wrapper guarantees only main-diagonal Z_2 covariance. It clamps the represented $x = 0$ and $y = 0$ boundaries and preserves the trained physical-frequency support during $16^2 \rightarrow 32^2$ transfer. For Navier–Stokes, static forcing is mapped to vorticity at $T = 5$ from zero initial vorticity. A trainable Stokes response supplies the linear forcing–diffusion solution, and dealiased bilinear blocks learn corrections associated with nonlinear convection.

3 Experiments

Protocol. All models minimize mean per-example relative L^2 error and are trained for 200 epochs with Adam, cosine learning-rate decay, gradient clipping, and three seeds. Lensing uses WDM

Table 1: Mean relative L^2 error on lensing. Training uses WDM; CDM and axion are zero-shot. Means are over three seeds; lower is better.

Model	$N_{\text{train}} = 800$			$N_{\text{train}} = 10,000$		
	WDM	CDM	Axion	WDM	CDM	Axion
BiC4-S3NO	.002531	.004079	.003159	.001748	.003117	.002298
FNO	.048764	.060053	.056793	.019929	.031484	.024623
PDNO	.148710	.171794	.302408	.006160	.010414	.010973
ALNO	.714950	.717425	.751121	.706509	.708961	.726390
LNO	.780616	.779103	.838612	.723844	.726720	.778324

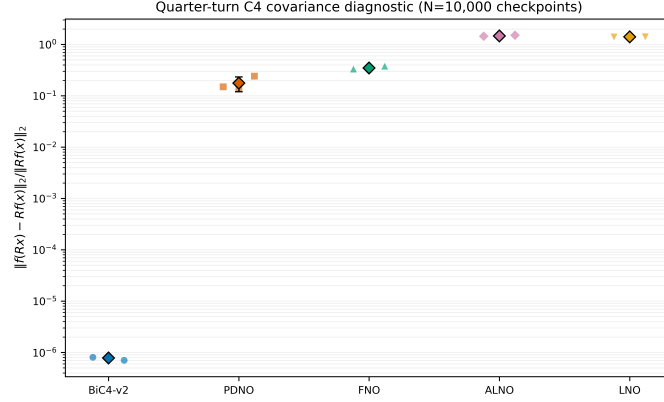


Figure 1: Quarter-turn covariance residual for validation-selected $N_{\text{train}} = 10,000$ lensing checkpoints.

training sets of 800 or 10,000 samples and evaluates held-out WDM plus zero-shot CDM and axion domains. Darcy uses 900 training and 100 validation examples at 16^2 , followed by native and zero-shot 32^2 evaluation. The Navier–Stokes development screen uses 900 training and 100 validation examples at 128^2 . FNO, PDNO, LNO, and ALNO are source-adapted controls trained through the same splits, preprocessing, objective, checkpoint selection, and evaluation pipeline; their values are not copied from the original publications.

Lensing and covariance. Table 1 shows that BiC4-S3NO has the lowest error in all six lensing settings. Increasing the WDM training set from 800 to 10,000 reduces its held-out WDM error from 0.002531 to 0.001748. Its CDM and axion performance indicates transfer beyond the training class. We separately evaluate the normalized quarter-turn residual $\mathcal{E}_{C_4} = \|\mathcal{F}(\rho x) - \rho \mathcal{F}(x)\|_2 / (\|\mathcal{F}(x)\|_2 + \epsilon)$. BiC4-S3NO preserves the prescribed transformation to numerical precision, whereas the comparison operators have no architectural guarantee (Figure 1).

Table 2: Mean relative L^2 error for PDE evaluations over three seeds. Darcy is trained at 16^2 ; “32” is zero-shot. NS is development validation.

Model	Darcy 16	Darcy 32	NS dev.
BiC4-S3NO	0.10428	0.18292	0.16145
FNO	0.14023	0.15780	0.40486
PDNO	0.08462	0.53387	0.59105
ALNO	0.36654	0.40441	0.56134
LNO	0.36855	0.44075	0.58670

PDE results and limitations. BiC4-S3NO ranks second on Darcy at both resolutions: PDNO leads at 16^2 , while FNO leads after transfer to 32^2 (Table 2). The half-open grid and sampled boundaries admit only the diagonal reflection used by the adapter, not a complete C_4 action. Moreover, the model has no analytic Darcy seed, and retaining the coarse-grid spectral support cannot reconstruct

unobserved fine-scale or boundary corrections. These effects have not been isolated by causal ablations. On Navier–Stokes development validation, BiC4-S3NO obtains the lowest mean error, 0.16145. The result is consistent with the Stokes seed modelling the linear response and the dealiased bilinear blocks representing unresolved nonlinear interactions, but it is not a held-out test benchmark.

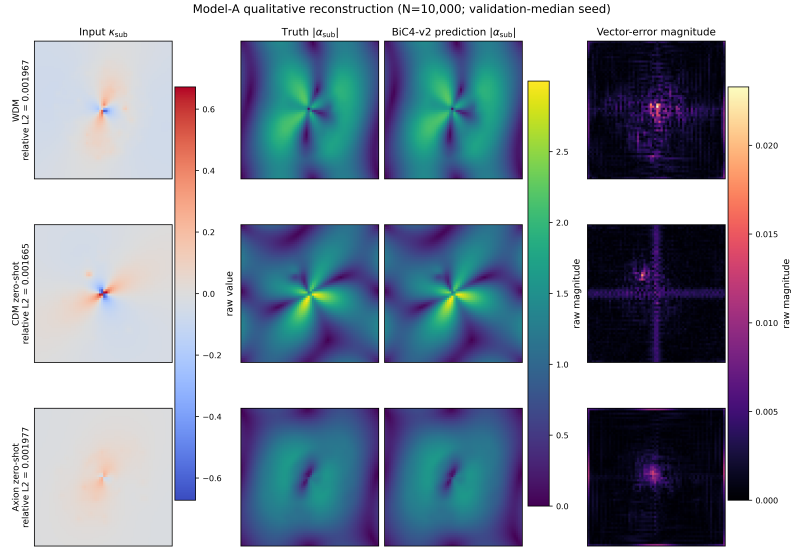


Figure 2: Representative validation-median BiC4-S3NO reconstructions for held-out WDM and zero-shot CDM and axion examples. Columns show convergence, true and predicted deflection magnitude, and vector-error magnitude.

4 Conclusion

BiC4-S3NO combines spin-typed spectral learning, symmetry-compatible bilinear interactions, and available analytic physics. It improves reconstruction and zero-shot dark-matter transfer while enforcing the intended lattice covariance, and performs strongly on the Navier–Stokes development task. Darcy demonstrates that symmetry should be matched to the actual discretization: incomplete grid actions and coarse supervision limit the benefit. These results support physics-seeded equivariant operators as compact scientific surrogates without implying universal superiority across PDEs.

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